



Mineral Physics Group
Research in Progress Meeting 2026
Wednesday 8th July



12:00-12:50

Registration

12:50-13:00

Introduction

13:00-13:40

Benjamin Hess (Invited)
University Of Oxford
Thermodynamics of Stressed Crystals: Stress–Composition Coupling and its Geological Consequences

13:40-14:00

Zhuoqi Lucas Li
Imperial College London
Mineral Boundary Plane Distributions in Basalt and Their Role in Geological Carbon Sequestration

14:00-14:20

Dongwoo Han
University Of Oxford
The Magnetic Structure of the Serpentinized Mantle Wedge: A Magnetic Fabric Study on the Sanbagawa Belt, Japan

14:20-14:40

Priyanka Pandit
Indian Institute Of Science Education And Research (IISER) Kolkata
Compositional Control on Hydrogen Incorporation and Deep Volatile Transport in Cold Slabs Across the Akimotoite–Bridgmanite Transition

14:40-15:00

Refreshment Break

15:00-15:40

Holger Stuniz (Invited)
University Of Tromsø, University Of Orléans
Recent Advances in Experimental Quartz Rheology and Consequences for the Plastic Deformation of Other Silicates

15:40-16:00

Peixue Dou
University College London
Point Defects as a Control on the Viscosity Jump in the Lower Mantle

16:00-16:20

Estelle Ledoux
University Of Oxford
The Stishovite to Post-Stishovite Phase Boundary Studied by Stress Cycling Experiments in a Ddac

16:20-16:40

Siddharth Khan
Laboratoire Magmas Et Volcans, CNRS
Aluminium Drives Hydrogen Incorporation in Pyrolitic Lower Mantle: Constraints from a Combined FTIR-ERDA-Nanosims Analysis

16:40-18:30

Posters and Refreshments

Sponsored by



ADVANCED
GEOSCIENCE
INSTRUMENTS
COMPANY



Magnetic
Measurements





Mineral Physics Group
Research in Progress Meeting 2026
Thursday 9th July



9:00-9:40

Banusha Kugabalan (Invited)
Imperial College London
Determining Palaeostress Signals Using Rock Magnetism.

9:40-10:00

Yi Li
University Of Oxford
Effects of Temperature-Driven Complexion Transitions and Redox Gradients on Grain-Boundary Plane Distributions in Olivine

10:00-10:20

Lei Liu
Institute Of Earthquake Forecasting, Chinese Earthquake Administration
Potential Significance of Iron-Bearing Superhydrous Phase B in the Formation of Mantle Low-Velocity Zones

10:20-10:40

Emily Wackan
University Of Glasgow
Comparisons Between the Trace Element Composition and Magnetic Structure of the Cloudy Zone in the Imilac and Odessa Iron-Nickel Meteorites

10:40-11:00

Brandon Fish
University Of Oxford
Nickel Content of Iron Meteorites

11:00-11:40

Refreshment Break

11:40-12:20

Zhi Li (Invited)
University College London
Predictive Modelling of Chemically Complex Earth-Forming Materials via Deep-Learning-Driven Atomistic Simulations.

12:20-12:40

Billy Rayner
University College London
Liquid Immiscibility in the Fe-O-S Ternary System at High Pressure and Temperature

12:40-13:00

Yuanjing Li
University College London
Light Element Exsolution In Fe-Si-C-(H) System in Planetary Liquid Cores

13:00-14:00

Break

14:00-15:00

Mineral Physics Group Annual General Meeting

Sponsored by



ADVANCED
GEOSCIENCE
INSTRUMENTS
COMPANY



Magnetic
Measurements

